BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Searching and Sorting Exam Practice

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

BLACK PEN: answer four questions before checking the answers.

1. How does merge sort combine two sorted lists?

2. What is algorithmic thinking?

A) Guessing a random order
B) Removing every input
C) Planning clear ordered steps
D) Drawing only screen layouts

3. "Compression is identical to encryption and primarily keeps data secret." Correct this. Explain why.

4. An app asks for pupils' locations and contacts without need. An online service uses only one language. Correct both claims. Explain why. (a) Posting all personal records improves privacy. (b) Technology has no effect on culture or language.

Today's objectives

Compare searching methods with correct prerequisites. Choose and trace a suitable sorting algorithm.

Key vocabulary - Read, cover, recall

Linear search

Checks array elements sequentially.

Binary search

Repeatedly halves a sorted search range.

Bubble sort

Adjacent comparisons and swaps.

Insertion sort

Inserts keys into a growing sorted section.

Write one key term and its meaning in your own words:

Do Now	I Do	We Do	You Do	Review & Improve
Recall previous learning	Read new knowledge	Practise with guidance	Work independently	Check my understanding

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Learn and practise

New learning - Key ideas

Choose a search method	Linear search inspects elements sequentially and works with unsorted lists. Binary search can discard halves but needs sorted input.
Trace before describing	When explaining an algorithm, show the actual array values, tested positions and changes-not merely the algorithm's name.
Three required sorting ideas	Bubble compares and swaps adjacent pairs; insertion places a key into a sorted portion; merge divides and combines sorted lists.
Answer what was asked	A question about one pass is not a request for complete sorting. A question about a search index uses zero-based array positions.

I Do - One worked example

A club has an unsorted list [5,2,9,1]. Explain which search method can be used immediately and trace the search for 9. [4 marks]

1. Because the list is unsorted, choose linear search [1].
2. Compare first item 5, not a match [1].
3. Compare second item 2, not a match [1].
4. Third item 9 matches, so return zero-based index 2 [1].

We Do - Guided questions

BLACK PEN: use the model, then explain your answer.

1. A sorted array is [2,4,6,8,10]. Trace binary search for 8. [4 marks]

2. A pupil sorts [4,1,3] using bubble sort. Show the first complete pass. [3 marks]

3. Explain why binary search should not be applied directly to [8,1,4,6] and how to prepare it. [3 marks]

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Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Complete the condition to print the position of 128 in this sorted array.

```
array drives = [32,64,128,256]
for i = 0 to 3
if ... then
print(i)
endif
next i
```

BLACK PEN: predict or complete the code; test, then improve.

For extended answers, continue on lined paper if needed.

You Do - Independent exam questions

Q2. State the prerequisite for binary search. [1 mark]

Q3. Trace the first bubble-sort pass on [5,3,4] and state the output array. [4 marks]

Q4. A dataset [6,2,9,1] is unsorted. Explain how to search for 9 without sorting, then describe how insertion sort would sort the whole list. [6 marks]

Exit Ticket - True or False?

Circle T or F before reviewing Page 4.

1. A binary search can safely discard half a list only when the relevant data is sorted. ☐ T ☐ F
2. Linear search inspects elements sequentially and works with unsorted lists ☐ T ☐ F
3. When explaining an algorithm, show the actual array values, tested positions and changes-not merely the ... ☐ T ☐ F

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: tick correct marking points, then correct errors.

Choose your hardest question to rewrite in full below.

Do Now - Answers

Q1 Compare the next item in each list. Take the smaller. Repeat, then add any items left over. ☐

Q2 C - Planning clear ordered steps ☐

Q3 It reduces the storage space required for data, using a suitable lossless or lossy method. ☐

Q4 (a) Sharing personal records can remove people's control over their data. (b) Technology can change language, access, traditions and who can take part. Consider benefits and harms, even when an action is legal. ☐

Code First - Model

```
array drives = [32,64,128,256]
for i = 0 to 3
if drives[i] == 128 then
print(i)
endif
next i
```

Output: 2

We Do - Check the model

1. First mid index 2 = 6 [1]. Target 8 is larger [1], so search right indices 3-4 [1]. Mid = 3 value 8 found at index 3 [1]. ☐

2. Compare 4 and 1 [1], swap to [1,4,3] [1]; compare 4 and 3, swap to [1,3,4] [1]. ☐

3. Array is unsorted [1]. Binary search discards halves based on order [1]. Sort ascending [1,4,6,8] before applying it [1]. ☐

Exit Ticket - Check

1. TRUE 2. TRUE 3. TRUE

You Do - Self-mark

Q2 The data must be sorted [1]. ☐

Q3 Compare 5>3 [1], swap to [3,5,4] [1]. Compare 5>4 [1], swap to [3,4,5] [1]. ☐

Q4 Linear search is suitable for unsorted data [1], comparing 6, 2 then 9 [1], found at index 2 [1]. Insertion sort builds sorted section [1]: insert 2 before 6 [1], then 9 and finally 1 into the correct places, giving [1,2,6,9] [1]. ☐

Try the question again

1. Copy the FULL question you found most difficult:

2. In GREEN PEN, write a complete new answer: